

YIFAN WANG

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SUMMARY

MSc Statistics student at Imperial College London focused on probabilistic time-series forecasting, uncertainty modelling, and deep learning. Experienced in time-series data processing, feature construction, model training and tuning, and metric-driven evaluation. Worked with multi-country renewable-energy data and evaluated probabilistic forecasting models across 7 public time-series datasets; seeking time-series forecasting and algorithm R&D internships.

EDUCATION

MSc Statistics (Data Science & Machine Learning), Imperial College London, UK 09/25 - 11/26

BSc (Hons) Mathematics, University of Liverpool, UK 09/23 - 06/25

First Class Honours (Top 2 in cohort)

BSc Applied Mathematics, Xi'an Jiaotong-Liverpool University, China 09/21 - 06/23

SKILLS

Programming & Tools Python (PyTorch, JAX, TensorFlow/Keras, NumPy, pandas), Git, LaTeX, mlx-lm-lora.

Statistics & ML Statistical inference, time-series modelling, feature engineering, deep sequence models (GRU/LSTM), model training, and hyperparameter tuning.

Probabilistic AI Flow matching, diffusion models, optimal transport, VAEs, Bayesian neural networks, uncertainty quantification, and calibration.

Model Evaluation CRPS, Energy Score, Recall/MRR/NDCG, overfitting diagnostics, and results interpretation.

LLMs Transformer internals, supervised fine-tuning, RLHF, GRPO, and DAPO.

Engineering Git, experimental scripting, reproducible results, and technical documentation.

Languages Chinese (Native); English (Proficient, IELTS 7.0, each part > 6.5).

RESEARCH EXPERIENCE

Visiting Research Student 07/26 - 09/26

Westlake University

China

- Investigating causal reasoning and interpretability in large language models, focusing on how SFT, GRPO, and DAPO post-training strategies affect causal reasoning behaviour.
- Fine-tuning Qwen2.5-0.5B with mlx-lm-lora for the DAMO project; reproducing [arXiv:2602.06337](#) related experiments and analysing how causal-inference post-training improves causal QA (ongoing).

SELECTED PROJECTS

GTFlow: A More Reliable Time-Series Forecaster [Website](#) 04/26 - 08/26

Imperial College London

UK

Enhancing flow matching for time-series distribution learning with Gaussian Process priors

- Traced Marcel Kollovich's research progression from TSDiff (NeurIPS 2023) to TSFlow (ICLR 2025), connecting score-based diffusion, flow matching, GP priors, and OT coupling to frame controlled ablations for the GTFlow extension.
- Addressed TSFlow's under-dispersion from learning with a single observed future by adding Gaussian Process posterior information as a trust-weighted flow-matching target.
- Evaluated CRPS and Energy Score on 7 GluonTS datasets with 3 random seeds; GTFlow reduced CRPS by 9.3%/13.0% and joint-trajectory Energy Score by 24.3%/17.6% on Solar/Taxi.
- Designed an L1 dispersion-ratio pre-training diagnostic for go/skip dataset decisions; all 21 seed-level decisions fell on the correct threshold side.

Stable Self-Distillation Flow Matching

07/25 - 09/25

Imperial College London

UK

Stabilising self-distillation flow matching with mini-batch optimal transport

- Implemented Eulerian and Lagrangian FMOT variants in JAX to study instability and gradient explosion in self-distillation flow matching, introducing mini-batch optimal transport coupling between source and target distributions.
- Evaluated 2D Moons/Gaussian $\rightarrow$ 8-Gaussians across 5 random seeds; E-FMOT achieved $W_2^2 = 0.724 \pm 0.149$ and NPE = 0.090 ± 0.040 while producing smoother transport paths with fewer training steps.

Session-Based Recommender System (YOOCHOOSE)

03/26 - 04/26

Imperial College London

UK

- Processed 21.45M click events, 6.01M sessions, and 49,499 items from YOOCHOOSE; created a final-24-hour temporal test split and built the session-cleaning, item-ID encoding, sequence-padding, and batching pipeline.
- Compared Embedding-GRU, LSTM, and Bi-GRU models in TensorFlow/Keras across learning rate, dropout, representation dimension, and batch size; the unidirectional GRU baseline achieved Recall@5=0.326, while Bi-GRU reached 0.741 offline, prompting analysis of possible future-context leakage.

AWARDS & SCHOLARSHIPS

- University of Liverpool:** Dean's List, Sampford Memorial, Willis & Brownlow Hill Prizes (top-of-cohort).
- Xi'an Jiaotong-Liverpool University:** Top 50 Excellence Award; Academic Achievement Award (Top 10%).